Efficiency is special calculation that can be done for a transformer, a transmission line (or power cable), or an entire power system. The general definition of efficiency of an input/output device like a transformer or transmission line is defined as the ratio in percent of the real power output of the device to the real power input of the device. This definition is shown mathematically below, where the Greek letter η (eta) is used.

$$\text{Device Efficiency} = \eta = \left(\frac{\text{real power output of the device}}{\text{real power input of the device}}\right)\times 100 = \left(\frac{P_{OUT}}{P_{IN}}\right)\times 100$$

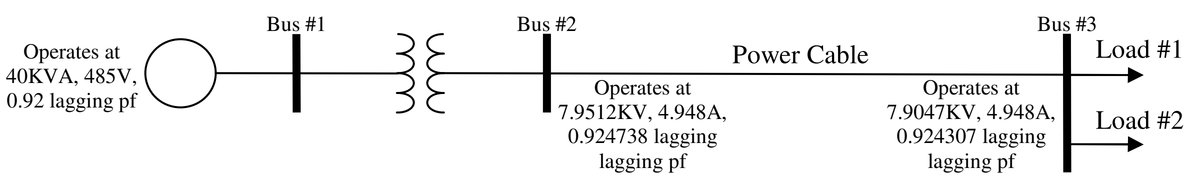
Efficiency of a transformer or transmission line is a measure in percent of what portion of the input real power actually makes it to the output. All real world transformers and transmission lines and machine systems have internal real power losses (heat, magnetic, vibration, etc.), and these losses prevent some of the input power from making it to the output. A real world transformer or transmission line is therefore never 100% efficient.

The efficiency of an entire power system is defined a bit differently. It is the ratio in percent of the net real power consumed by all loads of the system to the net real power supplied by all sources of the system. This definition is shown mathematically below.

$$\text{System Efficiency} = \eta_{sys} = \left(\frac{\text{net real power consumed by system loads}}{\text{net real power supplied by system sources}}\right)\times 100 = \left(\sum_j P_{Lj} \Big/ \sum_i P_{Si}\right)\times 100$$

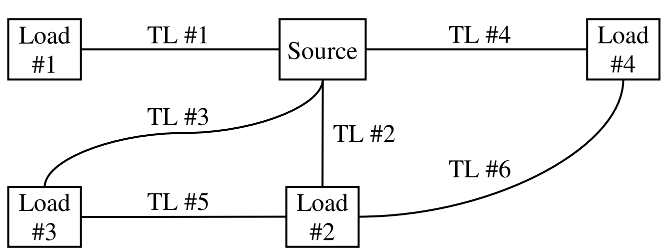
Efficiency of an entire power system is a measure in percent of what portion of the total real power supplied by the sources in the system actually makes it to the system loads. In between the sources and loads of a power system are the interconnect components (transformers and transmission lines) that we already stated have internal real power losses, and these losses prevent some of the source power from making it to the loads in a power system. A real world power system is therefore never 100% efficient. Typical efficiencies of power systems and power system components are in the 90-100% range, although it can be lower sometimes.

Example #1:
An analysis was done for the 1ϕ power system shown in the one line diagram below, and the results shown were obtained. Calculate the transformer efficiency, power cable efficiency, and power system efficiency.



$$V_1 = 485\,V,\quad S_1 = 40\ KVA,\quad pf_1 = 0.92\ \ lagging \qquad\rightarrow\qquad P_1 = S_1\cdot pf_1 = 36.8\ KW$$
$$V_2 = 7.9512\ KV,\quad I_2 = 4.948\ A,\quad pf_2 = 0.924738\ \ lagging \qquad\rightarrow\qquad P_2 = V_2 I_2\cdot pf_2 = 36.3815\ KW$$
$$V_3 = 7.9047\ KV,\quad I_3 = 4.948\ A,\quad pf_3 = 0.924307\ \ lagging \qquad\rightarrow\qquad P_3 = V_3 I_3\cdot pf_3 = 36.1519\ KW$$
$$\eta_{XFMR} = \left(\frac{P_2}{P_1}\right)\times 100 = 98.863\%,\qquad \eta_{PC} = \left(\frac{P_3}{P_2}\right)\times 100 = 99.369\%,\qquad \eta_{SYS} = \left(\frac{P_3}{P_1}\right)\times 100 = 98.239\%$$

Example #2:
The block diagram for a 3ϕ power system is given below.



Measurements were made in the system at the sources and loads, and the following table of information was recorded. Calculate the power system efficiency.

Source:	243.251V, 10.163A, 0.569821 lagging power factor
Load #1:	240.179V, consumes 529.712W and 631.286VAR
Load #2:	240.134V, consumes 720.804∠60°VA
Load #3:	238.892V, consumes 1141.396VA at a 0.866025 lagging power factor
Load #4:	242.881V, 2.698A, 0.342029 leading power factor

$$P_{3\phi S} = \sqrt{3} \cdot V_{LLS} I_{LS} \cdot pf_S = \sqrt{3}(243.251)(10.163)(0.569821) = 2439.920\,W$$

$$P_{3\phi L1} = 529.712\,W, \qquad P_{3\phi L2} = S_{3\phi L2}\cos\theta_{L2} = (720.804)\cos 60° = 360.402\,W$$

$$P_{3\phi L3} = S_{3\phi L3} pf_{L3} = (1141.396)(0.866025) = 988.477\,W$$

$$P_{3\phi L4} = \sqrt{3} \cdot V_{LL4} I_{L4} \cdot pf_{L4} = \sqrt{3}(242.881)(2.698)(0.342029) = 388.203\,W$$

$$\eta_{SYS} = \left(\frac{P_{Lnet}}{P_{Snet}}\right)\times 100 = \left(\frac{P_{3\phi L1} + P_{3\phi L2} + P_{3\phi L3} + P_{3\phi L4}}{P_{3\phi S}}\right)\times 100 = \left(\frac{2266.795\,W}{2439.920\,W}\right)\times 100 = 92.904\%$$

It is no accident that power system efficiencies are very high. Power system components are specifically designed to be highly efficient for economic reasons. In real world power systems, the sources of electrical energy are typically generators which are driven by mechanical devices (engines, turbines, etc.) that require some type of fuel to operate. Since this fuel must be purchased and has an associated cost in dollars, then the electrical energy produced by generators also has a related cost in dollars. In the operation of a power system, electrical energy is transferred from the sources to the loads through interconnect components. If electrical energy is lost in the interconnect components, then some of the dollars spent on fuel at the generators is being lost. If the efficiency of power system interconnect components can be improved, then less energy and consequently less dollars will be lost in the operation of the power system. This cost savings due to better system efficiency is illustrated in the example below.

Example #3:

In a stand alone power system, a diesel fuel operated generator supplies power to all of the loads in the system through a network of transformers and power cables. The average total continuous real power consumed at the system loads is 500KW. It can be assumed that the net cost of producing electrical energy at the generator is \$0.08/KWH (where KWH is kilowatt-hour). This net generator operating cost accounts for the diesel fuel cost and the combined efficiency of the generator with its mechanical drive. If the power system efficiency is 90%, then calculate the total dollar cost of the power lost in the system transformers and power cables in one year. Then repeat the calculation for a power system efficiency of 95%, and compare the results.

$$P_{Lnet} = 500\ KW, \qquad Cost = \$0.08/KWH$$

$$\eta_{sys} = \left(\frac{P_{Lnet}}{P_{Snet}}\right)\times 100 \quad \rightarrow \quad P_{Snet} = \frac{100 P_{Lnet}}{\eta_{sys}}$$

$$If\quad \eta_{sys} = 90\%\ \ then\ \ P_{Snet} = \frac{100 P_{Lnet}}{\eta_{sys}} = 555.5556\ KW\ \ and\ \ P_{Loss} = P_{Snet} - P_{Lnet} = 55.5556\ KW$$

$$Cost_{Loss} = (55.5556\ KW)(\$0.05/KWH)(24\ hr/day)(7\ day/wk)(52\ wk/yr) = \$38{,}826.67/yr$$

$$If\quad \eta_{sys} = 95\%\ \ then\ \ P_{Snet} = \frac{100 P_{Lnet}}{\eta_{sys}} = 526.3158\ KW\ \ and\ \ P_{Loss} = P_{Snet} - P_{Lnet} = 26.3158\ KW$$

$$Cost_{Loss} = (26.3158\ KW)(\$0.05/KWH)(24\ hr/day)(7\ day/wk)(52\ wk/yr) = \$18{,}391.58/yr$$

As can be seen in the analysis results, a cost savings of over \$20,000/yr can be realized if the efficiency is improved from 90% to 95% in this power system.